

Signals for today:

- Motivate complex exp. (input)
- Lead up to why we need the Laplace transform
- Properties of Laplace and how to calculate it

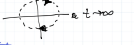
Thermal: $C\dot{x} + \frac{1}{L}x = q_{in}$

$q_{in} = \sum e^{st}$ where $t \in \mathbb{R}$
 $s \in \mathbb{C}$

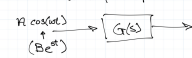
- 1) What is meant by a complex signal?
 - 2) What is the value of a complex exp?
- How can we input more realistic signals (e.g. steps, ramps, etc.)



$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$



- 3) Can we just input a single cosine into the system from a complex exponential?



4) $A e^{st} + A_2 e^{st} \rightarrow G(s) \rightarrow (A e^{st} + A_2 e^{st}) G(s)$
 $= A e^{st} G(s) + A_2 e^{st} G(s)$

Because of LTI property, a sum of complex exponentials is mapped linearly by the transfer function

$\sum_{n=0}^{\infty} A_n e^{st} \rightarrow G(s) \rightarrow \sum_{n=0}^{\infty} G(s) A_n e^{st}$

Back to #3: how convert $B e^{st}$ (r.r.m.m...) to $A \cos(\omega t)$

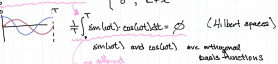
$B e^{st} \rightarrow B e^{(\sigma + j\omega)t}$
 $B [\cos(\omega t) + j\sin(\omega t)]$
 $B e^{j\omega t} + B e^{-j\omega t}$
 $B [\cos(\omega t) + j\sin(\omega t)] + B [\cos(\omega t) - j\sin(\omega t)]$
 $2B \cos(\omega t)$
Real!

We can convert an imaginary signal to a real signal by adding its complex conjugate.

- 5) We have cosines, but how do we get something more practical? (Fourier)
- Dense! Fourier Series allows us to create a very good square wave approx. w/ lots of complex exponentials

Brief review to Fourier Series

- 1) Important Property: orthogonality
- $\int_0^T e^{j k \omega t} e^{-j l \omega t} dt = \begin{cases} T & k=l \\ 0 & k \neq l \end{cases}$



2) $u(t) = \sum_{k=-\infty}^{\infty} U_k e^{j k \omega t}$ (Fourier Series)

$U_k = \frac{1}{T} \int_0^T u(t) e^{-j k \omega t} dt$

Homomorph of $e^{j\omega t}$ does $u(t)$ contain?

Signs, noise, through noise, cosine, ... something periodic

$u(t) = \begin{cases} 1 & 0 \leq t < T/2 \\ 0 & T/2 \leq t < T \end{cases}$ (repeated w/ period T)

$U_0 = \frac{1}{2}$ $U_k = \frac{1}{j\pi k}$ (k odd) $U_k = 0$ (k even)

$u(t) = \frac{1}{2} + \frac{2}{\pi} \left(\sin(\omega t) + \frac{1}{3} \sin(3\omega t) + \frac{1}{5} \sin(5\omega t) + \dots \right)$

$u(t) =$ all terms up to ∞ harmonics

- 6) How do we find the response of the spring-mass-damper to something like a square wave?

- 6a) How do we find the response (cos) to a cosine?



Important: what allows to input many comp. exp. into the system is the LTI prop. of the system and not necessarily the signal.

Signal: $A e^{st}$

System: $G(s)$

Output: $A G(s) e^{st}$

Complex number can be written in polar form $a + jb = m e^{j\phi}$

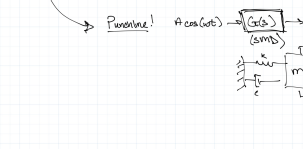
m is the magnitude ($a^2 + b^2$)

ϕ is the phase

Parseval! $A \cos(\omega t) \rightarrow G(s) \rightarrow A \left[\frac{G(s)}{2} \cos(\omega t) + \frac{G(s^*)}{2} \cos(\omega t) \right]$

$G(s) = \frac{1}{ms^2 + cs + k}$

$G(j\omega) = \frac{1}{-m\omega^2 + j c \omega + k}$



$F = 3 \cos(5t)$

$x = 3 \left[\frac{G(j5)}{2} \cos(5t) + \frac{G(-j5)}{2} \cos(5t) \right]$

$G(s) = \frac{1}{ms^2 + cs + k}$

$G(j\omega) = \frac{1}{-m\omega^2 + j c \omega + k}$

$G(-j\omega) = \frac{1}{-m\omega^2 - j c \omega + k}$

$G(s) = \frac{1}{ms^2 + cs + k}$

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